

Module 4: Energy, reproduction and populations

4.1 Energy, metabolism and exercise

This section builds on the knowledge and understanding gained in Module 2 by considering the role of ATP as an immediate source of energy for biological processes and the biochemical pathways of cellular respiration that generate ATP.

The concept of aerobic fitness is explored in the context of the effects of exercise on the respiratory and cardiovascular systems. The effects of exercise on skeletal muscles and the mechanism of muscle contraction at cellular level are also considered.

The process of cellular respiration requires an efficient supply of oxygen and the importance of respiratory pigments as oxygen carriers is illustrated by reference to haemoglobin and myoglobin. Learners also gain an insight into the unique properties of fetal haemoglobin in meeting the metabolic needs of the fetus during development.

Learners are expected to apply knowledge, understanding and other skills gained in this section to new situations and/or to solve related problems.